

Physical Anchors for the ppBHPT–NR Amplitude and Time Rescalings

Draft note for discussion

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1 Motivation

The goal is to replace as many freely fitted parameters as possible in the generalized ppBHPT–NR scaling model by analytically motivated structures. The starting point is the calibration used in BHPTNRSur1dq1e4,

$$h_{\ell m}^{\text{cal}}(\tau; q) = \alpha_{\ell}(q) h_{\ell m}^{\text{pp}}(t; q), \quad \tau = \beta(q)t, \quad (1)$$

where the original model fits a common time-rescaling parameter $\beta(q)$ and mode-dependent amplitude parameters $\alpha_{\ell}(q)$ to NR-hybrid waveforms. The fitted functions were represented as quartic polynomials in $1/q$ and constrained to approach unity as $q \rightarrow \infty$ [1]. A complementary lesson comes from higher-harmonic template-bank construction: low-order PN relations can be applied to the full nonlinear (2, 2) mode to provide useful approximations to the (3, 3) and (4, 4) modes well beyond the strict early-inspiral regime [2].

Our current agentic generalization promotes these constants to time-dependent functions. Schematically,

$$\alpha(t) = \alpha_i + (1 - S)\alpha_L \widehat{\Delta p} + S(\alpha_E \Delta E + \alpha_J \Delta J), \quad (2)$$

$$\beta(t) = (1 - S)(\beta_i + \beta_L \Delta p) + S\beta_r. \quad (3)$$

This is flexible, but it introduces strong degeneracies. The purpose of this note is to identify leading-order post-Newtonian (PN) predictions that can be imposed as anchors before allowing the agent to fit residual corrections.

2 Conventions

Let

$$q = \frac{m_1}{m_2} \geq 1, \quad M = m_1 + m_2, \quad X_1 = \frac{m_1}{M} = \frac{q}{1+q}, \quad X_2 = \frac{m_2}{M} = \frac{1}{1+q}, \quad (4)$$

and

$$\nu = X_1 X_2 = \frac{q}{(1+q)^2}, \quad \eta_{\text{pp}} = \frac{m_2}{m_1} = \frac{1}{q}, \quad \Delta = X_1 - X_2 = \frac{q-1}{q+1}. \quad (5)$$

Here η_{pp} is the perturbative mass ratio used in the point-particle calculation; it should not be confused with the symmetric mass ratio ν .

We use the standard PN frequency variable

$$x = (M\Omega)^{2/3}, \quad (6)$$

with the understanding that the ppBHPT and comparable-mass waveforms initially use different mass units. Equation (1) translates between the two dimensionless time coordinates.

3 Origin of the 6/5 power law

3.1 Matching the Newtonian chirp rate

At leading order, the adiabatic inspiral satisfies

$$\frac{dx_{\text{pp}}}{dt} = \frac{64}{5}\eta_{\text{pp}}x_{\text{pp}}^5, \quad \frac{dx_{\text{NR}}}{d\tau} = \frac{64}{5}\nu x_{\text{NR}}^5. \quad (7)$$

Under the time map $\tau = \beta t$, equality of the orbital phase implies

$$\Omega_{\text{NR}} = \frac{\Omega_{\text{pp}}}{\beta}, \quad x_{\text{NR}} = \beta^{-2/3}x_{\text{pp}}. \quad (8)$$

Using Eqs. (7) and (8), the transformed ppBHPT chirp rate becomes

$$\frac{dx_{\text{NR}}}{d\tau} = \frac{64}{5}\eta_{\text{pp}}\beta^{5/3}x_{\text{NR}}^5. \quad (9)$$

Matching this to the comparable-mass result gives

$$\eta_{\text{pp}}\beta^{5/3} = \nu. \quad (10)$$

Since

$$\frac{\nu}{\eta_{\text{pp}}} = \frac{m_1^2}{M^2} = X_1^2, \quad (11)$$

we obtain

$$\boxed{\beta_{\text{LO}}(q) = \left(\frac{\nu}{\eta_{\text{pp}}}\right)^{3/5} = X_1^{6/5} = \left(\frac{q}{1+q}\right)^{6/5}.} \quad (12)$$

Thus, the exponent 6/5 is not an empirical numerical coincidence. It is the product of two facts: the Newtonian chirp rate scales linearly with the relevant mass ratio, and x transforms as $\beta^{-2/3}$ under a time rescaling. The resulting coefficient is proportional to $\beta^{5/3}$, whose inversion produces the exponent 3/5; multiplying by X_1^2 then gives $X_1^{6/5}$.

3.2 Matching the leading quadrupole amplitude

At Newtonian order,

$$h_{22}^{\text{pp,N}} \propto \eta_{\text{pp}}x_{\text{pp}}, \quad h_{22}^{\text{NR,N}} \propto \nu x_{\text{NR}}. \quad (13)$$

The amplitude map requires

$$\alpha_2\eta_{\text{pp}}x_{\text{pp}} = \nu x_{\text{NR}}. \quad (14)$$

Using $x_{\text{pp}} = \beta^{2/3}x_{\text{NR}}$ gives

$$\alpha_{2,\text{LO}} = \frac{\nu}{\eta_{\text{pp}}}\beta^{-2/3} = X_1^2\beta^{-2/3}. \quad (15)$$

Substituting Eq. (12),

$$\boxed{\alpha_{2,\text{LO}} = X_1^{6/5} = \beta_{\text{LO}}.} \quad (16)$$

Therefore the equality $\alpha_2 = \beta$ follows from simultaneously matching the leading chirp rate and the leading (2,2)-mode amplitude.

The original BHPTNRSur paper motivated the simpler mass-unit scaling $\alpha = \beta = X_1$, and then optimized the constants against NR. Numerically, however, its fitted $\alpha_2(q)$ and $\beta(q)$ are much closer to $X_1^{6/5}$ over the calibration interval $3 \leq q \leq 10$.

4 Leading-PN prediction for the higher harmonics

The leading Newtonian multipoles for a nonspinning circular binary have the mass-ratio structure

$$h_{\ell m}^{\text{N}} \propto \nu c_{\ell+\epsilon_{\ell m}}(\nu) x^{(\ell+\epsilon_{\ell m})/2}, \quad (17)$$

where $\epsilon_{\ell m} = (\ell + m) \bmod 2$ and

$$c_k(\nu) = X_2^{k-1} + (-1)^m X_1^{k-1}. \quad (18)$$

The paper determines α_ℓ using the $\ell = m$ modes. For these modes, $\epsilon_{\ell\ell} = 0$. Because the fitted α_ℓ rescales the amplitude, we define

$$C_\ell \equiv \left| X_2^{\ell-1} + (-1)^\ell X_1^{\ell-1} \right|. \quad (19)$$

The test-particle waveform has the corresponding leading behavior

$$h_{\ell\ell}^{\text{pp,N}} \propto \eta_{\text{pp}} x_{\text{pp}}^{\ell/2}. \quad (20)$$

Applying the same time map as above, the amplitude-matching condition becomes

$$\alpha_\ell \eta_{\text{pp}} \left(\beta^{2/3} x_{\text{NR}} \right)^{\ell/2} = \nu C_\ell x_{\text{NR}}^{\ell/2}. \quad (21)$$

Hence

$$\boxed{\alpha_{\ell,\text{LO}} = X_1^2 C_\ell \beta^{-\ell/3}}. \quad (22)$$

Using $\beta = X_1^{6/5}$ gives

$$\boxed{\alpha_{\ell,\text{LO}} = C_\ell X_1^{2-2\ell/5}}. \quad (23)$$

For the modes calibrated in the paper,

$$\alpha_{2,\text{LO}} = X_1^{6/5}, \quad (24)$$

$$\alpha_{3,\text{LO}} = \Delta X_1^{4/5}, \quad (25)$$

$$\alpha_{4,\text{LO}} = (1 - 3\nu) X_1^{2/5}, \quad (26)$$

$$\alpha_{5,\text{LO}} = \Delta(1 - 2\nu). \quad (27)$$

These formulas explain two qualitative features of the fitted calibration:

1. the higher-mode amplitude factors decrease more strongly at comparable mass ratios; and
2. the suppression becomes progressively stronger with increasing ℓ , because the leading multipoles contain additional powers of the unequal-mass factors C_ℓ .

An equivalent expression directly in q is

$$\boxed{\alpha_{\ell,\text{LO}}(q) = \left[1 + (-1)^\ell q^{1-\ell} \right] (1 + q^{-1})^{-(1+3\ell/5)}, \quad (\ell = m, \ q > 1)}. \quad (28)$$

5 Comparison with the fitted coefficients in the paper

The paper fits

$$\alpha_\ell(q) = 1 + \frac{A_\ell}{q} + \frac{B_\ell}{q^2} + \frac{C_\ell^{\text{fit}}}{q^3} + \frac{D_\ell}{q^4}. \quad (29)$$

Expanding Eq. (28) in $u = 1/q$ gives

$$\alpha_{2,\text{LO}} = 1 - \frac{6}{5}u + \frac{33}{25}u^2 - \frac{176}{125}u^3 + \frac{924}{625}u^4 + \mathcal{O}(u^5), \quad (30)$$

$$\alpha_{3,\text{LO}} = 1 - \frac{14}{5}u + \frac{108}{25}u^2 - \frac{714}{125}u^3 + \frac{4389}{625}u^4 + \mathcal{O}(u^5), \quad (31)$$

$$\alpha_{4,\text{LO}} = 1 - \frac{17}{5}u + \frac{187}{25}u^2 - \frac{1558}{125}u^3 + \frac{11339}{625}u^4 + \mathcal{O}(u^5), \quad (32)$$

$$\alpha_{5,\text{LO}} = 1 - 4u + 10u^2 - 20u^3 + 34u^4 + \mathcal{O}(u^5). \quad (33)$$

The leading $1/q$ coefficients compare as follows:

ℓ	A_ℓ from BHPTNRSur	$A_{\ell,\text{LO}}^{\text{PN}}$
2	-1.330	-1.200
3	-3.067	-2.800
4	-3.909	-3.400
5	-4.509	-4.000

The monotonic trend and most of the magnitude are already explained by the leading-PN mass-ratio factors. This is particularly notable because the fitted coefficients were obtained from waveform-level calibration rather than imposed from PN theory.

A direct comparison of the full functions is shown below. “Fit” denotes Eq. (29) with the published coefficients, while “LO” denotes Eq. (27).

q	α_2		α_3		α_4		α_5	
	LO	Fit	LO	Fit	LO	Fit	LO	Fit
3	0.7081	0.7087	0.3972	0.3826	0.3899	0.3195	0.3125	0.1919
5	0.8035	0.8044	0.5762	0.5671	0.5423	0.4932	0.4815	0.3802
8	0.8682	0.8661	0.7078	0.6963	0.6713	0.6323	0.6241	0.5407

The leading-order result is excellent for $\ell = 2$, accurate at the few-percent level for $\ell = 3$, and captures a substantial fraction of the trend for $\ell = 4, 5$. The fitted amplitudes for $\ell = 4, 5$ are systematically below the leading-order prediction, especially near $q = 3$. Plausible sources include higher-PN amplitude terms, strong-field and merger-ringdown effects, the finite size of the secondary relative to the point-particle approximation, and the use of a single constant α_ℓ over a long inspiral-merger-ringdown interval.

One should not over-interpret agreement between the individual polynomial coefficients beyond the leading term. The published quartic is a finite-interval regression over $3 \leq q \leq 10$, not an asymptotic expansion measured arbitrarily close to $q = \infty$. Its higher-order coefficients can therefore absorb higher-PN physics and compensate one another.

6 Using the full nonlinear (2, 2) mode as a higher-mode backbone

6.1 Motivation from higher-harmonic template banks

Reference [2] uses the stationary-phase approximation to relate the Fourier-domain higher harmonics to the full nonlinear (2, 2) waveform. At leading PN order, the intrinsic mode amplitudes have the schematic relations

$$A_{33}(f) \propto \Delta V_3(f) A_{22} \left(\frac{2f}{3} \right), \quad (34)$$

$$A_{44}(f) \propto (1 - 3\nu) V_4(f)^2 A_{22} \left(\frac{f}{2} \right), \quad (35)$$

where $V_m(f) = (2\pi M f / m)^{1/3}$ and the omitted numerical constants depend on the mode and waveform conventions. The corresponding leading phase relation is

$$\Psi_{\ell m}(f) \simeq \frac{m}{2} \Psi_{22} \left(\frac{2f}{m} \right) + \delta_{\ell m}^{(0)}. \quad (36)$$

The important point is that A_{22} and Ψ_{22} in these expressions need not be truncated PN waveforms: one can insert the full inspiral–merger–ringdown (2, 2) mode and use only the *inter-mode transfer factor* from leading PN theory. Figure 6 of Ref. [2] shows that the resulting amplitude approximation remains useful close to merger. The same paper also finds that Eq. (36) is not sufficiently accurate on its own: mode-dependent phase residuals can reach a few radians and must be modeled separately. Moreover, its final search-bank implementation uses bank-averaged normalized higher-mode amplitudes because this gives a small additional improvement in the nonlinear regime. These observations motivate a physics baseline plus a low-dimensional residual, rather than imposing the PN relation as an exact identity.

6.2 Time-domain backbone map

Write the calibrated quadrupole as

$$h_{22}^{\text{cal}}(\tau) = A_{22}^{\text{cal}}(\tau) e^{-i\phi_{22}^{\text{cal}}(\tau)}, \quad \tau = \tau(t), \quad \frac{d\tau}{dt} = \beta(t, q). \quad (37)$$

A time-domain analogue of Eqs. (35)–(36) is

$$\mathcal{T}_{\ell m}[h_{22}] = \mathcal{N}_{\ell m} C_{\ell m}(\nu) x^{(\ell + \epsilon_{\ell m} - 2)/2} A_{22} \exp \left[-i \frac{m}{2} \phi_{22} + i \delta_{\ell m}^{(0)} \right], \quad (38)$$

where $\mathcal{N}_{\ell m}$ is a convention-dependent numerical constant and

$$C_{\ell m}(\nu) = \left| X_2^{\ell + \epsilon_{\ell m} - 1} + (-1)^m X_1^{\ell + \epsilon_{\ell m} - 1} \right|, \quad C_{\ell m} \rightarrow 1 \quad (q \rightarrow \infty). \quad (39)$$

For the diagonal modes considered in the BHPTNRSur calibration,

$$C_{33} = \Delta, \quad C_{44} = 1 - 3\nu, \quad C_{55} = \Delta(1 - 2\nu). \quad (40)$$

Directly using $\mathcal{T}_{\ell m}[h_{22}^{\text{cal}}]$ as the predicted higher mode is a useful diagnostic, but it discards mode-specific nonlinear information already present in ppBHPT, including plunge excitation, quasinormal-mode content, and possible mode mixing. A safer construction is therefore to transfer only the *quadrupole calibration* onto the existing ppBHPT higher mode.

6.3 Ratio-transfer construction for ppBHPT–NR calibration

At the amplitude level, define

$$A_{\ell m}^{\text{pred}}(\tau) = A_{\ell m}^{\text{pp}}(t) \frac{A_{22}^{\text{cal}}(\tau)}{A_{22}^{\text{pp}}(t)} C_{\ell m}(\nu) \left(\frac{x_{\text{NR}}(\tau)}{x_{\text{pp}}(t)} \right)^{(\ell + \epsilon_{\ell m} - 2)/2} \rho_{\ell m}^A(\tau, q), \quad (41)$$

where $\rho_{\ell m}^A \rightarrow 1$ in the point-particle limit. Since

$$\frac{x_{\text{NR}}}{x_{\text{pp}}} = \beta^{-2/3}, \quad (42)$$

Eq. (41) becomes

$$\boxed{\alpha_{\ell m}^{\text{backbone}}(t, q) = \alpha_{22}(t, q) C_{\ell m}(\nu) \beta(t, q)^{-(\ell + \epsilon_{\ell m} - 2)/3} \rho_{\ell m}^A(t, q).} \quad (43)$$

For the diagonal modes,

$$\alpha_{33}^{\text{backbone}} = \alpha_{22} \Delta \beta^{-1/3} \rho_{33}^A, \quad (44)$$

$$\alpha_{44}^{\text{backbone}} = \alpha_{22} (1 - 3\nu) \beta^{-2/3} \rho_{44}^A, \quad (45)$$

$$\alpha_{55}^{\text{backbone}} = \alpha_{22} \Delta (1 - 2\nu) \beta^{-1} \rho_{55}^A. \quad (46)$$

If $\alpha_{22} = \beta = X_1^{6/5}$ and $\rho_{\ell\ell}^A = 1$, these reduce exactly to Eq. (27). Thus the earlier leading-PN formulas can be reinterpreted as the special case of a more general construction: retain the full nonlinear quadrupole calibration and transfer it to each higher mode using the leading-PN inter-mode factors.

A corresponding phase transfer is

$$\boxed{\phi_{\ell m}^{\text{pred}}(\tau) = \phi_{\ell m}^{\text{pp}}(t) + \frac{m}{2} \left[\phi_{22}^{\text{cal}}(\tau) - \phi_{22}^{\text{pp}}(t) \right] + \delta\phi_{\ell m}(\tau, q).} \quad (47)$$

The residual $\delta\phi_{\ell m}$ is essential: the higher-harmonic template-bank study shows that the naive phase transfer does not remain accurate into the nonlinear regime. In the present calibration, a common time map already transfers much of the phase evolution, so Eq. (47) should be viewed as the correct form if an additional quadrupole phase correction is introduced.

6.4 Comparison with the fitted leading $1/q$ coefficients

The backbone construction can be tested without setting the quadrupole calibration to its pure leading-order value. Let the published fits begin as

$$\alpha_2 = 1 + \frac{A_2}{q} + \dots, \quad \beta = 1 + \frac{A_\beta}{q} + \dots, \quad (48)$$

with $A_2 = -1.330$ and $A_\beta = -1.238$ from Ref. [1]. Setting the residuals in Eq. (46) to unity gives

$$A_{33}^{\text{backbone}} = A_2 - 2 - \frac{A_\beta}{3} \simeq -2.917, \quad (49)$$

$$A_{44}^{\text{backbone}} = A_2 - 3 - \frac{2A_\beta}{3} \simeq -3.505, \quad (50)$$

$$A_{55}^{\text{backbone}} = A_2 - 4 - A_\beta \simeq -4.092. \quad (51)$$

These values can be compared directly with the independently fitted coefficients:

mode	independent fit	22-backbone prediction	fractional difference
33	−3.067	−2.917	4.9%
44	−3.909	−3.505	10.3%
55	−4.509	−4.092	9.2%

Therefore, once the fitted nonlinear 22 amplitude and common time rescaling are transferred using the leading-PN mass factors, most of the leading higher-mode calibration is already explained. The remaining residual is small enough to motivate a constrained agentic search for $\rho_{\ell m}^A$ rather than an independent unconstrained $\alpha_\ell(q)$ for each mode.

6.5 Recommended residual model

A compact inspiral model is

$$\rho_{\ell m}^A(x, q) = \exp \left[\nu \left(r_1^{\ell m} x + r_{3/2}^{\ell m} x^{3/2} + r_2^{\ell m} x^2 \right) + \mathcal{O}(\nu^2 x) \right], \quad (52)$$

which enforces positivity and the point-particle limit. For a full inspiral–merger–ringdown model, one can use

$$\ln \rho_{\ell m}^A = (1 - S) \ln \rho_{\ell m}^{A, \text{insp}} + S \ln \rho_{\ell m}^{A, \text{MR}}, \quad (53)$$

where the merger–ringdown residual is organized using remnant mass, spin, and quasinormal-mode information. Off-diagonal modes, especially modes affected by spherical–spheroidal mixing, should be tested separately rather than assumed to follow the same residual hierarchy as $\ell = m$ modes.

7 What does PN theory say about α_L ?

The parameter α_L is specific to the current time-dependent model in Eq. (3); it does not appear in the original BHPTNRSur calibration. Its numerical value depends on the definition of p_{loss} , the waveform start time, the fitted transition point p_0 , the switch S , and how corrections are distributed among α_L , β_L , α_E , and α_J . It is therefore not a convention-independent PN coefficient.

At Newtonian order the amplitude scaling is constant, so

$$\boxed{\alpha_L^{\text{Newtonian}} = 0.} \quad (54)$$

A nonzero fitted α_L is naturally interpreted as a projection of higher-PN and strong-field amplitude evolution onto the chosen linear drift variable.

For the (2, 2) mode, the standard 1PN amplitude contains

$$h_{22} = h_{22}^{\text{N}} \left[1 + \left(-\frac{107}{42} + \frac{55}{42} \nu \right) x + 2\pi x^{3/2} + \dots \right]. \quad (55)$$

Relative to the test-particle limit, the leading finite-mass correction at fixed x is therefore

$$\frac{h_{22}^{\text{comparable}}}{h_{22}^{\text{test particle}}} \simeq 1 + \frac{55}{42} \nu x + \dots. \quad (56)$$

This suggests replacing the ad hoc linear drift by a PN-organized inspiral factor,

$$\alpha_{22}^{\text{insp}}(x, q) = X_1^{6/5} \left[1 + \frac{55}{42} \nu x + a_{3/2}(\nu) x^{3/2} + a_2(\nu) x^2 + \dots \right], \quad (57)$$

with only residual coefficients left for calibration. Mapping Eq. (57) onto the current $\widehat{\Delta p}$ basis would produce an effective α_L , but the result would depend on the selected interval and on the time-rescaling model. In particular, β_L changes the frequency sampled at a given mapped time and is therefore degenerate with an amplitude drift.

Numerically, our present per- q fits show that the early-inspiral combination $\alpha_i + \alpha_L$ is very close to $X_1^{6/5}$ for $q \gtrsim 4$. This is consistent with interpreting $X_1^{6/5}$ as the leading physical anchor and α_L as a small higher-order correction rather than as an independent baseline parameter.

8 Suggested reduced parameterization

The preferred hierarchy is to fit the quadrupole calibration first and transfer it to the higher modes:

$$\beta_{\text{insp}}(x, q) = X_1^{6/5} [1 + \delta_\beta(x, \nu)], \quad (58)$$

$$\alpha_{22}^{\text{insp}}(x, q) = X_1^{6/5} [1 + \delta_{22}(x, \nu)], \quad (59)$$

$$\alpha_{\ell m}^{\text{insp}}(x, q) = \alpha_{22}^{\text{insp}}(x, q) C_{\ell m}(\nu) \beta_{\text{insp}}(x, q)^{-(\ell + \epsilon_{\ell m} - 2)/3} \rho_{\ell m}^A(x, q). \quad (60)$$

All residuals should satisfy

$$\delta_\beta, \delta_{22}, \ln \rho_{\ell m}^A \rightarrow 0 \quad \text{as} \quad \nu \rightarrow 0. \quad (61)$$

A PN-like basis is

$$\delta(x, \nu) = \nu \left(c_1 x + c_{3/2} x^{3/2} + c_2 x^2 + \dots \right) + \nu^2 \left(d_1 x + d_{3/2} x^{3/2} + \dots \right), \quad (62)$$

rather than independent low-order polynomials for every phenomenological parameter and mode.

For merger and ringdown, a separate physical anchor is probably required. The natural scales are the remnant mass, remnant spin, quasinormal-mode frequencies, and excitation amplitudes; one should not expect the Newtonian chirp-mass or inter-mode scaling to remain exact. The ppBHPT higher modes should nevertheless be retained as the nonlinear backbone, with only a switched residual calibrated to NR.

9 Concrete tasks

A practical sequence for the next iteration is:

1. Reproduce Eqs. (12)–(27) directly from the Newtonian PN modes and verify conventions against the ppBHPT waveform normalization.
2. Plot the published $\alpha_\ell(q)$ and $\beta(q)$ together with both the pure leading-order curves and the 22-backbone predictions of Eq. (46) over $3 \leq q \leq 10$.
3. Implement two diagnostics: (i) direct higher modes generated from $\mathcal{T}_{\ell m}[h_{22}^{\text{cal}}]$, and (ii) the preferred ratio-transfer construction in Eqs. (41) and (47). Compare inspiral-only and full IMR errors separately.
4. Refit the current model after imposing $\beta_i = X_1^{6/5}$ and either $\alpha_i + \alpha_L = X_1^{6/5}$ or an explicitly PN-organized $\alpha_{22}^{\text{insp}}(x, q)$.

5. For each higher mode, set the analytic backbone to Eq. (43) and ask the agent to model only $\ln \rho_{\ell m}^A$ and $\delta\phi_{\ell m}$ using a small PN-like basis. Begin with 33, then test 44 and 55; treat off-diagonal modes separately.
6. Compare validation error, extrapolation to $q = 2$, and the parameter-correlation matrix before and after imposing these anchors. A modest loss in in-sample accuracy may be worthwhile if extrapolation and identifiability improve substantially.
7. Perform ablations that set $\rho_{\ell m}^A = 1$ and $\delta\phi_{\ell m} = 0$ separately. This will measure how much of the improvement comes from the nonlinear 22 transfer, the leading-PN mass factors, and the learned residuals.

10 Main takeaway

The leading ppBHPT–NR time and amplitude rescalings contain substantial analytically predictable structure:

$$\boxed{\beta_{\text{LO}} = X_1^{6/5}, \quad \alpha_{2,\text{LO}} = X_1^{6/5}, \quad \alpha_{\ell m}^{\text{backbone}} = \alpha_{22} C_{\ell m}(\nu) \beta^{-(\ell+\epsilon_{\ell m}-2)/3} \rho_{\ell m}^A.} \quad (63)$$

For the quadrupole, the 6/5 exponent follows from simultaneous matching of the Newtonian chirp rate and amplitude. For higher harmonics, the leading-PN inter-mode factors explain most of the hierarchy in the fitted α_ℓ coefficients. The nonlinear (2, 2) mode should therefore serve as a common calibration backbone, while the existing ppBHPT higher modes retain their mode-specific strong-field and ringdown structure. The agent then needs to discover only small amplitude and phase residuals, rather than rediscovering an independent calibration for every mode from scratch.

References

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